

GIS Data Converter – Mini Project Plan

Objective

Build a web-based geospatial data conversion service. Users upload files in one format, configure conversion options (e.g., target format, coordinate system), and download the processed results. (*Feel free to use any AI coding assistant for the mini project*)

Deliverables:

1. Backend (Server-Side)

Core Functionalities (Mandatory):

- **File Upload API:** Accept file uploads via multipart/form-data with support for multiple geospatial formats.
- **Task Management:**
 - Generate a unique task ID for each conversion job.
 - Maintain and expose task status (**pending**, **processing**, **completed**, **failed**).
 - Provide an endpoint to fetch conversion results.
 - Provide a download endpoint for the output file.
- **Conversion Engine** – Must implement the **Top 5 priority conversions**:
 1. **GeoJSON ↔ CSV** – Export annotations with properties as spreadsheets; import features from tabular data with coordinates.
 2. **GeoTIFF → COG** (Cloud-Optimized GeoTIFF) – Optimize rasters for fast cloud tile serving.
 3. **COG / Raster → GeoJSON** – Vectorize raster features (e.g., polygonization of binary masks).
 4. **GeoJSON → Raster** – Rasterize vector features into a grid (for machine learning datasets).
 5. **Reprojection** – Convert spatial data between different EPSG coordinate reference systems (e.g., WGS84 to Web Mercator).
- **Data Validation:**
 - Validate uploaded file formats and extensions.
 - Validate GeoJSON structure (RFC 7946 compliance).
 - Validate GeoTIFF integrity and read coordinate system metadata.
- **API Documentation:** Provide machine-readable API documentation (e.g., OpenAPI/Swagger).

Optional / Advanced Features (Bonus):

- *Implement any of the following 5 secondary conversions:*
 - *GeoJSON ↔ Shapefile (ESRI standard).*
 - *GeoJSON ↔ GeoPackage (.gpkg).*
 - *GeoJSON ↔ KML/KMZ (Google Earth).*

- *Multi-band GeoTIFF → Individual single-band COGs (e.g., extract RGB/NIR from Sentinel-2).*
- *GeoJSON ↔ COCO JSON (ML annotation format interoperability).*
- **Persistence Layer** (Database optional):
 - *Store task metadata (ID, filename, source/target formats, timestamp, status, user context).*
 - *Provide conversion history endpoints.*
 - *(Optional) Simple user management (registration, login, API keys).*
- **Asynchronous Processing:** *Offload long-running conversions to a background worker queue with real-time progress updates (e.g., via WebSocket or Server-Sent Events).*

2. Frontend (Client-Side / UI)

Core Functionalities (Mandatory):

- **Upload Interface:**
 - Drag-and-drop or click-to-select file upload.
 - Visual feedback for accepted file types and sizes.
- **Configuration Panel:**
 - Dropdown selectors for input format and desired output format.
 - Coordinate system selector (EPSG code input or dropdown).
 - Basic parameter controls (e.g., rasterization resolution, band selection).
- **Progress & Status Dashboard:**
 - Real-time display of the current task status.
 - Visual progress indicator (e.g., progress bar) while processing.
- **Results Section:**
 - Prominent download button upon completion.
 - Display output file name and size.
- **Error Handling:** Clear user-friendly error messages for invalid files or failed conversions.

Optional / Advanced Features (Bonus):

- **Interactive Map Preview:**
 - *Visualize the resulting GeoJSON or COG on an embedded map (e.g., using a WebGL-based map library).*
 - *Display attribute tables for vector data (CSV/GeoJSON preview).*
 - **Batch Upload:** *Allow multiple files to be uploaded and converted simultaneously.*
 - **History Panel:**
 - *List previous conversion tasks.*
 - *Allow re-downloading past results without re-uploading.*
 - **Advanced Parameters:** *Expose professional GDAL-like options (compression method, NoData values, output resolution, tiling).*
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Sample Datasets

To test the conversion engine, provide the candidate with the following real-world, publicly available datasets. *(Feel free to explore and find your own dataset.)*

Satellite Imagery / GeoTIFF

Dataset	Description	Source
OSGeo GeoTIFF Samples	Official standard test images for GeoTIFF.	https://download.osgeo.org/geotiff/samples/
GeoTIFF Benchmark Files	A packaged set of multiple GeoTIFF test files.	wget https://s3.us-east-2.amazonaws.com/geotiff-benchmark-sample-files/geotiff_sample_files.tar.gz

Vector Data (GeoJSON / Shapefile)

Dataset	Description	Source
World Countries Boundary	Simple global country boundaries in GeoJSON.	https://github.com/python-visualization/folium/blob/master/examples/data/us-states.json (US states)
GADM Global Administrative Areas	Worldwide administrative boundaries (available in Shapefile, KML, GeoPackage).	https://gadm.org/data.html

Deliverables Summary Checklist

BACKEND (Server-Side)

- File Upload API (multi-format support)
- Task Management (ID, status, result fetch, download)
- Conversion Engine (Mandatory Top 5: GeoJSON<->CSV, GeoTIFF->COG, Raster->GeoJSON, GeoJSON->Raster, Reprojection)
- File Format Validation (GeoTIFF, GeoJSON, etc.)
- API Documentation
- *(Optional) Secondary Conversions (at least 1 of the 5 listed)*

- *(Optional) Database for task history*
- *(Optional) Async processing with progress updates*

FRONTEND (Client-Side UI)

- File Upload Interface (drag-and-drop)
- Conversion Configuration Panel (format selection, EPSG, parameters)
- Real-time Progress Display
- Result Download Button
- User-friendly Error Messages
- *(Optional) Map Preview for output data*
- *(Optional) Conversion History Panel*
- *(Optional) Batch file support*

DELIVERED WITH:

- Clear setup/installation instructions (no language specifics required)
- API endpoint examples (using curl/Postman)
- Demonstration using the provided sample datasets

Feel free to choose any programming language, web framework, database, or UI library.